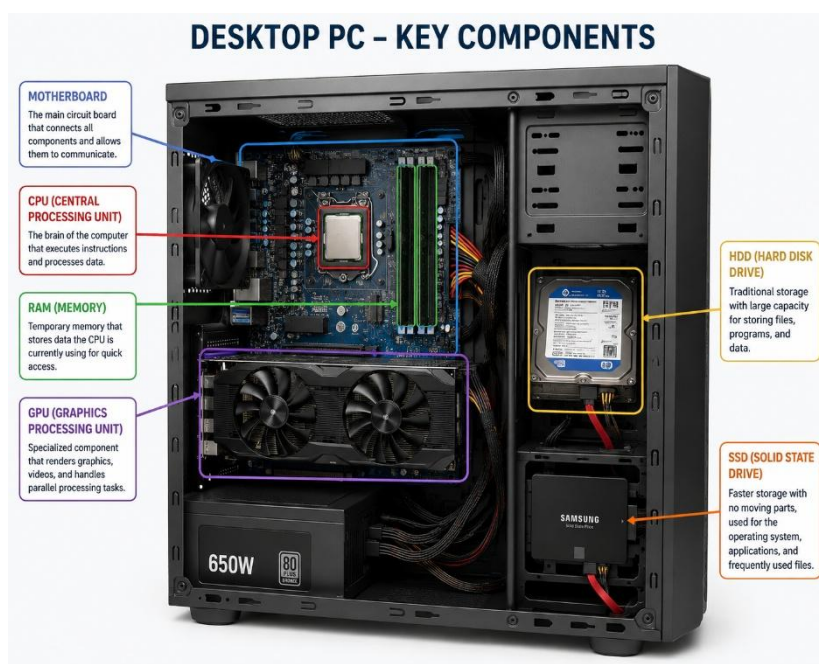


ASSIGNMENT:1

HARDWARE COMPONENTS

Q1. Draw a labelled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.

ANS.



**Q2. Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.

Hint: Think about which one is faster, which is more shock-resistant, and where you would usually find each type (laptop, gaming PC, etc.).**

ANS.

Feature	HDD (Hard Disk Drive)	SSD (Solid State Drive)
Speed	Slower read/write speed	Much faster read/write speed
Durability	Less durable because it has moving parts	More durable and shock-resistant because it has no moving parts
Storage Capacity	Larger storage at a lower cost	Smaller storage is usually more expensive
Power Consumption	Uses more power	Uses less power, improving battery life
Noise	Produces noise due to spinning disks	Silent operation

Q3. Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

ANS.

- **CPU (Central Processing Unit):-**

- The CPU processes game instructions, controls player actions, and manages game logic. A faster CPU helps the game run smoothly with fewer delays.

- **RAM (Random Access Memory):**
 - RAM temporarily stores game data that the CPU needs while the game is running.
- **GPU (Graphics Processing Unit):-**
 - The GPU is responsible for rendering graphics, animations, and visual effects in the game.

Q4. Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

ANS.

1. **CPU (Central Processing Unit):** Upgrade the CPU to improve game performance, reduce lag, and handle game calculations more efficiently.
 2. **GPU (Graphics Processing Unit):** Upgrade the GPU to achieve higher frame rates, better graphics quality, and smoother gameplay.
 3. **RAM (Random Access Memory):** Increase RAM to at least 16 GB so the game can run smoothly while allowing other applications to run in the background.
 4. **SSD (Solid State Drive):** Install the game on an SSD to reduce loading times and improve overall system responsiveness.
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